Project Set-up - Electrical

- **Project Name**
- **Location**
- **Design Stage – Drop down (CD, SD, DD, TD, IFC)**
- **Revision**
- **Client Name**
- **Plot ref**
- **Project Category – (residential, hotel, mixed-use, office, mall, hospital, school, industrial, aircraft hangars, others)**
- **Plot load allocation (kW/kVA – choice of units)**
- **Ambient air temperature**
- **Ambient soil temperature**
- **Soil Thermal Resistivity degC.m/W**
- **Altitude**
- **No of Floors – (No of basements – user input;**
  GF – user input
  No of podium levels – user input
  No of typical floors – user input
  Mechanical floor-1 ref – user input
  Mechanical floor-2 ref – user input
  Mechanical floor-3 ref – user input)

Calculation set-up

- **Region - Drop-down -> "Dubai", "Abudhabi", others;**
- **Upstream short-circuit capacity – user input (eg. 350MVA, 500MVA)**
- **Calculations – Checkboxes -> "Current Carrying Capacity", "Voltage drop", "Short-circuit", "Let through energy", "Earth Fault Loop Impedance"**

Demand factors (category-wise)

1. Lighting; - xx
2. Power;- xx
3. AC;-xx
4. Heater;-xx
5. Laundry;- xx
6. Kitchen;-
7. Hand Dryer;-
8. Pumps;-
9. Fire Pumps;-
10. Special Equipment;-
11. Lifts & Equipment;-
12. Spare-

Demand factors (category-wise) – for Dubai alone

Residential- xx;
Landlord- xx;
Office- xx;
Retail- xx;

LV Panel – Dubai Format

PROJECT :-DUBAI CREEK GARDENS

PLANNED COMPLETION DATE -

LV PANEL-P4-B2-04 mirage

DETAILS OF CONNECTED LOAD, MAX. DEMAND & kWHr METERING

LOCATION OF LV PANEL: LV ROOM-B2- GF

CIRCUIT / FEEDER / SMDB / DB NO.	SP/TP	ACB/MCCB ISOLATOR RATING (AMPS)	FAULT DUTY kA	CABLE SIZE, TYPE AND NO. OF CORES: NO. OF CORES 1C/2C/4C	TYPE XLPE/PVC/SWA	SIZE	ECC SIZE 1C, mm²	LENGTH OF CABLES (Mtrs.)	CONNECTED LOAD (kW): R-PH kW	Y-PH kW	B-PH kW	Total Connected / Installed Load (TCL) * kW	Max. Demand / Operational Load (MDL) kW	Proposed Type & No. of kWHr Meters: 1-PH (1)	3-PH (2)	* LV/H (3)	REMARKS
INCOMER	4P	1600	50	2*1C PER PH +1*1C FOR NEUTRAL	XLPE	630	2*150	15	386.14	323.56	349.66	1059.36	711.83				
Mobile Generator Provision	4P	1600	50														
OUTGOING WAYS:																	
PFCC-P4-B2-04 - 200KVAR	TP	400	50	1*4C	XLPE	300	150	15									
SMDB-P4-B2-L7-APT-01	TP	250	50	1*4C	XLPE	240	120	131	43.12	35.09	38.50	116.71	75.86		9		
SMDB-P4-B2-L7-APT-02	TP	300	50	1*4C	XLPE	300	150	131	47.55	38.37	42.13	128.05	83.23		10		
SMDB-P4-B2-L8-APT-01	TP	250	50	1*4C	XLPE	300	150	135	43.12	35.09	38.50	116.71	75.86		9		
SMDB-P4-B2-L8-APT-02	TP	300	50	1*4C	XLPE	300	150	135	47.55	38.37	42.13	128.05	83.23		10		
SMDB-P4-B2-L9-APT-01	TP	250	50	1*4C	XLPE	300	150	139	43.12	35.09	38.50	116.71	75.86		10		
SMDB-P4-B2-L9-APT-02	TP	300	50	1*4C	XLPE	300	150	139	47.55	38.37	42.13	128.05	83.23		10		
SMDB-P4-B2-L10-APT-01	TP	350	50	1*4C	XLPE	300	150	143	57.48	46.51	51.11	155.10	100.81		12		
SMDB-POOL-P4-B2-01	TP	400	50	1*4C	XLPE	300	150	50	51.67	51.67	51.67	155.00	124.00				
SPARE	TP	125	50						5.00	5.00	5.00	15.00	9.75				
HARMONIC FILTER- LV4	TP	100	50														
MDB CONNECTED TO: DEWA LV DB/TRANSFORMER: TR-P4-B2-04								TOTAL CONNECTED LOAD PER PHA	####	####	####	1059.36	711.83	-	70	-	TOTAL

DEMAND FACTOR:_____ 0.67 MAX. DEMAND: ---------- 711.83 kW * OVERALL TOTAL CONNECTED / INSTALLED LOAD (TCL) : 1059.36 kW TOTAL BUILD-UP AREA: .....

CONSULTANT / CONTRACTOR: ..... TEL:..... FAX :.....

* TCL - shall include all loads proposed to be installed including standby, spare and future load provisions.

Type of Meter (Rating of Incomer) (1) Upto 60A (1 Phase) (2) Upto 125A (3 Phase) (3) LV CT ____/____A / HV CT ____/ ____.A (* 200/5Amps CT Metering)

Page 1

Load Schedule

LOAD ANALYSIS

	LOAD DESCRIPTION	CONNECTED LOAD W: R	T	B	DEMAND FACTOR	MAXIMUM DEMAND LOAD W: R	T	B
1	LIGHTING	4425	4813	2358		2876.25	3128.45	1532.7
2	SOCKET OUTLETS	23000	15000	14500		14950	9750	9425
3	HVAC	700	1400	2700		455	910	1755
4	WATER HEATER		9900	18000			6435	11700
5	LAUNDRY EQUIPMENT		7500	10500			4875	6825
6	KITCHEN EQUIPMENT	28850	7100	3050		18752.5	4615	1982.5
7	HAND DRYER							
8	PUMPS							
9	FIRE PUMPS							
10	SPECIAL EQUIPMENT		800				520	
11	LIFts and Escalators							
12	Spare	500				325		
	TOTAL	57475	46513	51108		37359	30233	33220
			155.10 KW				100.81 KW	

Load Summary

Manual input

Transformer, SMDB, MDB – Dubai Format

PROJECT :-DUBAI CREEK GARDENS

PLANNED COMPLETION DATE :

SMDB-P4-B2-L10-APT-01

DETAILS OF CONNECTED LOAD, MAX. DEMAND & kWHr METERING

mirage

LOCATION OF SMDB: ELECTRICAL ROOM - B2-L10

CIRCUIT FEEDER SMDB / DB NO.	WAY NO	SP/TP	ACB/MCCB ISOLATOR RATING (AMPS)	FAULT DUTY kA	CABLE SIZE, TYPE AND NO. OF CORES: NO. OF CORES 1C/2C/4C	TYPE XLPE/PVC/SWA	SIZE	ECC SIZE 1C, mm²	LENGTH OF CABLES (Mtrs.)	CONNECTED LOAD (kW): R-PH kW	Y-PH kW	B-PH kW	Total Connected / Installed Load (TCL) * kW	Max. Demand / Operational Load (MDL) kW	Proposed Type & No. of kWHr Meters: 1-PH (1)	3-PH (2)	* LV/H (3)	REMARKS
INCOMER		TP	**350**	**50**	1*4C	XLPE	300	150	143	**57.48**	**46.51**	**51.11**	**155.10**	**100.81**				
OUTGOING WAYS:																		
DB-P4-B1-2BR-L10-01	1RYB	TP	40	35	1*4C	XLPE	16	16	31	4.96	4.07	4.49	13.53	8.79		1		
DB-P4-B1-2BR-L10-02	2RYB	TP	40	35	1*4C	XLPE	25	16	44	4.96	4.07	4.49	13.53	8.79		1		
DB-P4-B1-3BR-L10-03	3RYB	TP	40	35	1*4C	XLPE	35	25	53	6.47	4.80	5.73	16.99	11.05		1		
DB-P4-B1-3BR-L10-04	4RYB	TP	40	35	1*4C	XLPE	35	25	65	6.47	4.80	5.73	16.99	11.05		1		
DB-P4-B1-1BR-L10-05	5RYB	TP	40	35	1*4C	XLPE	25	16	60	3.95	3.31	3.44	10.70	6.95		1		
DB-P4-B1-1BR-L10-06	6RYB	TP	40	35	1*4C	XLPE	25	16	57	3.95	3.31	3.44	10.70	6.95		1		
DB-P4-B1-2BR-L10-07	7RYB	TP	40	35	1*4C	XLPE	25	16	49	4.96	4.07	4.49	13.53	8.79		1		
DB-P4-B1-1BR-L10-08	8RYB	TP	40	35	1*4C	XLPE	25	16	40	3.95	3.31	3.44	10.70	6.95		1		
DB-P4-B1-1BR-L10-09	9RYB	TP	40	35	1*4C	XLPE	16	16	31	3.95	3.31	3.44	10.70	6.95		1		
DB-P4-B1--1BR-L10-10	10RYB	TP	40	35	1*4C	XLPE	16	16	23	3.95	3.31	3.44	10.70	6.95		1		
DB-P4-B1-2BR-L10-11	11RYB	TP	40	35	1*4C	XLPE	16	16	20	4.96	4.07	4.49	13.53	8.79		1		
DB-P4-B1-2BR-L10-12	12RYB	TP	40	35	1*4C	XLPE	10	10	20	4.96	4.07	4.49	13.53	8.79		1		
MDB CONNECTED TO: DEWA LV DB/TRANSFORMER: TR-P4-B2-04					TOTAL CONNECTED LOAD PER PH.					57.475	46.513	51.108	155.10	100.81	-	12	-	TOTAL
SMDB CONNECTED TO: LV PANEL-P4-B2-04																		

DEMAND FACTOR:_____ 0.65 MAX. DEMAND: ---------- 100.81 kW * OVERALL TOTAL CONNECTED / INSTALLED LOAD (TCL) : 155.10 kW TOTAL BUILD-UP AREA: .....

CONSULTANT / CONTRACTOR: ..... TEL:..... FAX :.....

* TCL - shall include all loads proposed to be installed including standby, spare and future load provisions.

Type of Meter (Rating of Incomer): (1) Upto 60A (1 Phase) (2) Upto 125A (3 Phase) (3) LV CT ___/___A / HV CT ___/ ___.A (* 200/5Amps CT Metering)

Page 1

Load Schedule

LOAD ANALYSIS

	LOAD DESCRIPTION	ECTED LOAD W R	T	B	DEMAND FACTOR	HAND LOAD W R	T	B
1	LIGHTING	4425	4813	2358		2876.25	3128.45	1532.7
2	SOCKET OUTLETS	23000	15000	14500		14950	9750	9425
3	HVAC	700	1400	2700		455	910	1755
4	WATER HEATER		9900	18000			6435	11700
5	LAUNDRY EQUIPMENT		7500	10500			4875	6825
6	KITCHEN EQUIPMENT	28850	7100	3050		18752.5	4615	1982.5
7	HAND DRYER							
8	PUMPS							
9	FIRE PUMPS							
10	SPECIAL EQUIPMENT		800				520	
11	LIFts and Escalators							
12	Spare	500				325		
	TOTAL	57475	46513	51108		37359	30233	33220
			155.10 KW				100.81 KW	

Load Summary

- **For transformers** – Primary voltage, secondary voltage, kVA rating, %z, %R, %X fields are required in addition to above details;
- **For SMDB's, MDB's** – Double incomer may be applicable in certain cases

Busbar – Dubai Format

PROJECT :-AVARRA BY PALACE

mirage

DETAILS OF CONNECTED LOAD, MAX. DEMAND & kWHr METERING

PREPARED BY :SK REV: TENDER

CHECKED BY : NS

BUSBAR (BB1-T-AP01)-1250A

LOCATION OF SMDB: ELECTRICAL ROOM-L1

CIRCUIT / FEEDER / SMDB / DB NO.	WAY NO	SP/TP		ACB/MCCB / ISOLATOR / RATING (AMPS)	FAULT DUTY kA	CABLE SIZE, TYPE AND NO. OF CORES: NO. OF CORES 1C/2C/4C	TYPE XLPE/PVC/SWA	SIZE	ECC SIZE 1C, mm²	LENGTH OF CABLES (Mtrs.)	CONNECTED LOAD (kW): R-PH kW	Y-PH kW	B-PH kW	Total Connected / Installed Load (TCL) * kW	Max. Demand / Operational Load (MDL) kW	Proposed Type & No. of kWHr Meters: 1-PH (1)	3-PH (2)	* LV/HV (3)	REMARKS
											295.28	288.62	301.50	885.40	531.24				
BUSBAR (BB1-T-AP01)-1250A TPN+E																			
SMDB-T-L1-AP	1RYB	TP		100	35	1 * 4C	XLPE	35	25	4	19.99	19.66	20.45	60.10	36.06				
SMDB-T-L2-AP	2RYB	TP		125	35	1 * 4C	XLPE	50	25	6	24.99	24.46	25.55	75.00	45.00				
SMDB-T-L3-AP	3RYB	TP		125	35	1 * 4C	XLPE	50	25	6	24.99	24.46	25.55	75.00	45.00				
SMDB-T-L4-AP	4RYB	TP		125	35	1 * 4C	XLPE	50	25	6	24.99	24.46	25.55	75.00	45.00				
SMDB-T-L5-AP	5RYB	TP		125	35	1 * 4C	XLPE	50	25	6	24.99	24.46	25.55	75.00	45.00				
SMDB-T-L6-AP	6RYB	TP		125	35	1 * 4C	XLPE	50	25	6	24.99	24.46	25.55	75.00	45.00				
SMDB-T-L7-AP	7RYB	TP		125	35	1 * 4C	XLPE	50	25	6	24.99	24.46	25.55	75.00	45.00				
SMDB-T-L8-AP	8RYB	TP		125	35	1 * 4C	XLPE	50	25	6	24.99	24.46	25.55	75.00	45.00				
SMDB-T-L9-AP	9RYB	TP		125	35	1 * 4C	XLPE	50	25	6	24.99	24.46	25.55	75.00	45.00				
SMDB-T-L10-AP	10RYB	TP		125	35	1 * 4C	XLPE	50	25	6	24.99	24.46	25.55	75.00	45.00				
SMDB-T-L11-AP	11RYB	TP		100	35	1 * 4C	XLPE	35	25	6	19.33	18.73	19.65	57.71	34.63				
SMDB-T-L12-AP	12RYB	TP		160	35	1 * 4C	XLPE	70	35	6	31.08	30.09	31.45	92.62	55.57				
MDB CONNECTED TO: DEWA LV DB/TRANSFORMER: TR-1							TOTAL CONNECTED LOAD PER PHASE				295.28	288.62	301.50	885.40	531.24	-	-	-	TOTAL
BUSBAR CONNECTED TO: LV PANEL-1																			

DEMAND FACTOR:..... 0.60 MAX. DEMAND: ---------- 531.24 kW * OVERALL TOTAL CONNECTED / INSTALLED LOAD (TCL) : 885.40 kW TOTAL BUILD-UP AREA: .....

CONSULTANT / CONTRACTOR: ..... TEL:..... FAX :.....

* TCL - shall include all loads proposed to be installed including standby, spare and future load provisions.

Metering ?

Type of Meter (Rating of Incomer): (1) Upto 60A (1 Phase) (2) Upto 125A (3 Phase) (3) LV CT ...../.....A / HV CT ...../ .....A (* 200/5Amps CT Metering)

SHEET 1 OF 1

Load Schedule

LOAD ANALYSIS

	LOAD DESCRIPTION	ECTED LOAD W R	T	B	DEMAND FACTOR	HAND LOAD W R	T	B
1	LIGHTING	4425	4813	2358		2876.25	3128.45	1532.7
2	SOCKET OUTLETS	23000	15000	14500		14950	9750	9425
3	HVAC	700	1400	2700		455	910	1755
4	WATER HEATER		9900	18000			6435	11700
5	LAUNDRY EQUIPMENT		7500	10500			4875	6825
6	KITCHEN EQUIPMENT	28850	7100	3050		18752.5	4615	1982.5
7	HAND DRYER							
8	PUMPS							
9	FIRE PUMPS							
10	SPECIAL EQUIPMENT		800				520	
11	LIFts and Escalators							
12	Spare	500				325		
	TOTAL	57475	46513	51108		37359	30233	33220
			155.10 KW				100.81 KW	

Load Summary

- **For Busbars – Ampere rating field – manual or calculated**
- **For Busbars – R, X & Z (90deg C) fields should be available; these values can be taken from multiple brands, or manual options should be available for various Ratings**

Power Factor Capacitor Bank

PROJECT :-DUBAI CREEK GARDENS

AREA : AL JADDAF, DUBAI

mirage

POWER FACTOR CORRECTION CAPACITOR BANK SIZING

LOCATION OF PFCC: LV ROOM-B2-GF

PFCC-P4-B2-05 - 400KVAR **PFC-G-01**

CONNECTED LOAD DETAILS OF MDB-1	TDL IN KW	LOAD POWER FACTOR			CORRECTED POWER FACTOR			CAPACITIVE KVAR REQUIRED PER KW	TOTAL CAPACITIVE KVAR REQUIRED	REMARKS
		COS Φ1	Φ1	TAN Φ1	COS Φ2	Φ2	TAN Φ2			
LIGHTING	66.004	0.90	25.842	0.484	0.93	21.565	0.395	0.089	5.881	
SOCKET OUTLETS	113.280	0.80	36.870	0.750	0.93	21.565	0.395	0.355	40.189	
HVAC	498.16	0.80	36.870	0.750	0.93	21.565	0.395	0.355	176.736	
WATER HEATER	6.24	0.90	25.842	0.484	0.93	21.565	0.395	0.089	0.556	
LAUNDRY EQUIPMENT		0.80	36.870	0.750	0.93	21.565	0.395	0.355		
KITCHEN EQUIPMENT	4.40	0.85	31.788	0.620	0.93	21.565	0.395	0.225	0.988	
HAND DRYER										
PUMPS	40.00	0.85	31.788	0.620	0.93	21.565	0.395	0.225	8.981	
FIRE PUMPS										
SPECIAL EQUIPMENT- (GSM AND ELV EQUIPMENT)	213.31	0.85	31.788	0.620	0.93	21.565	0.395	0.225	47.893	
LIFT	122.40	0.85	31.788	0.620	0.93	21.565	0.395	0.225	27.481	
SPARE	1.60	0.80	36.870	0.750	0.93	21.565	0.395	0.355	0.566	
TOTAL CONNECTED LOAD IN KW	**1065**	**KW**			REQUIRED RATING OF CAPACITOR BANK				**309.27**	**KVAR**
					SELECTED RATING OF CAPACITOR BANK				**400**	**KVAR**

Page 1

Cables

- **Cross section area (CSA) field – manual or calculated**
- **Type of insulation (PVC/XLPE/Fire rated)**
- **Type of sheath (PVC/LSZH)**
- **Method of installation**
- **Depth of laying (for buried cables)**
- **No of cables installed together (for grouping factor)**
- **R, X & Z (90deg C) fields should be available; these values can be taken from multiple brands, or manual options should be available for various sizes**
- **R, X & Z (20deg C) fields should be available; these values can be taken from multiple brands, or manual options should be available for various sizes**

LV Panel, MDB, SMDB – Abudhabi Format

Project/ Building : BVLGARI MANSION ABU DHABI MANSIONS & ESTATES | Client:EAGLE HILLS REGIONAL PROPERTIES | mirage

Address/Location : Massnova Island, Abu Dhabi | Building: ESTATE A | Consultant: mirage

MDB ref: MDB-6EA-B1-01 | Incoming cable size/type: x mm2 /// +*1C mm2 / ECC

Location: ELECTRICAL RM-BASEMENT | Main Breaker Type &Rating: AT/AF | Date: July 2026

Fed from: LV PANEL-6EA-GF-01 | Document No: Mir-BVL-Elec-01

		Single/ Three Phase	Rating (Amps)		Cable Type	CABLE SIZE, TYPE AND NO. OF CORES				CONNECTED LOAD			Total	Diversity Type	Diversity factor	Diversified Load			Total	REMARKS
			RCD (mA)	MCCB / ACB		No. of Cables	No of cores	Size mm²	ECC mm²	R-PH (kW)	Y-PH (kW)	B-PH (kW)	(kW)			R-PH (kW)	Y-PH (kW)	B-PH (kW)	(kW)	
	OUTGOING WAYS:																			
1	SMDB-6EA-B1-CM-01		#VALUE!	AT/AF	//					#VALUE!	#VALUE!	#VALUE!	#VALUE!	######	#VALUE!	#VALUE!	#VALUE!	#VALUE!	#VALUE!	
2	SMDB-6EA-B1-WS11		#VALUE!	AT/AF	//					19.53	18.03	17.53	55.10	######	#VALUE!	#VALUE!	#VALUE!	#VALUE!	#VALUE!	6.50kW STANDBY
3	SMDB-6EA-GF-KIT1		#VALUE!	AT/AF	//					#VALUE!	#VALUE!	#VALUE!	#VALUE!	######	#VALUE!	#VALUE!	#VALUE!	#VALUE!	#VALUE!	
4	MCC-6EA-B1-ME11		#VALUE!	AT/AF	//					15.98	15.98	15.98	47.94	######	#VALUE!	#VALUE!	#VALUE!	#VALUE!	#VALUE!	2.75kW STANDBY
5	POOL MAIN PANEL		ELR	AT/AF	//			95		67.67	67.67	67.67	203.00	MO	0.50	33.83	33.83	33.83	101.50	ELR
6	SMDB-6EA-B1-CM-02		#VALUE!	160AT/AF	//			95		#VALUE!	#VALUE!	#VALUE!	#VALUE!	SM	#VALUE!	#VALUE!	#VALUE!	#VALUE!	#VALUE!	
7	SPARE			AT/100AF						0.83	0.83	0.83	2.50	PWR	0.50	0.42	0.42	0.42	1.25	
8	SPARE			AT/100AF						0.83	0.83	0.83	2.50	PWR	0.50	0.42	0.42	0.42	1.25	
	Total Connected Phase Loads																			

Gross Area	4032	Diversified Load types	PWR	LTG	WH	MO	A/C	Total connected Load (Kw):	#VALUE!
		MDB Load	#VALUE!	#VALUE!	#####	######	#VALUE!	Load After Diversity	#VALUE!
		W/sq.m	#VALUE!	#VALUE!	#####	######	#VALUE!	Overall Diversity:	#VALUE!

Page 1

Load Schedule

LOAD ANALYSIS

	LOAD DESCRIPTION	ECTED LOAD W			DEMAND	HAND LOAD W		
		R	T	B	FACTOR	R	T	B
1	LIGHTING	4425	4813	2358		2876.25	3128.45	1532.7
2	SOCKET OUTLETS	23000	15000	14500		14950	9750	9425
3	HVAC	700	1400	2700		455	910	1755
4	WATER HEATER		9900	18000			6435	11700
5	LAUNDRY EQUIPMENT		7500	10500			4875	6825
6	KITCHEN EQUIPMENT	28850	7100	3050		18752.5	4615	1982.5
7	HAND DRYER							
8	PUMPS							
9	FIRE PUMPS							
10	SPECIAL EQUIPMENT		800				520	
11	LIFts and Escalators							
12	Spare	500				325		
	TOTAL	57475	46513	51108		37359	30233	33220
			155.10 KW				100.81 KW	

Load Summary

- **Where multiple incomers (fed from) applies, the source names shall be separated by “/”; For eg: LV Panel/Generator**

Generator Load Analysis

mirage

AVARRA BY PALACE

REVIEWER	DATE	PROJECT No.
		MLD_
CHECKER	DATE	CALC. REF.
ORIGINATOR	DATE	REVISION
		A

LOAD ANALYSIS - GENERATORS

FROM	TO	CONNECTED LOAD IN KW			TOTAL LOAD KW	DEMAND FACTOR-NORMAL	DEMAND FACTOR-CASE-1	DEMAND FACTOR-CASE-2	NORMAL OPERATION	CASE-1 POWER FAILURE	CASE-2 FIRE CONDITION	REMARKS
		R-PH KW	Y-PH KW	B-PH KW								
EMDB-GF-LL						0.8						
EMDB-GF-LL												
EMDB-GF-01	ESMDB-L76-LL-01											
ESMDB-L76-LL-01	GSM ISOLATOR-01-GEM RM - ROOF	5.500	5.500	5.500	16.500	0.800	0.800	0.800	13.200	13.200	13.200	
ESMDB-L76-LL-01	EMCC-L76-LL-01											
EMCC-L76-LL-01	STAIRCASE-3 PRESSURISATION FAN-TWR-LR-SPF-07	2.500	2.500	2.500	7.500	0.600	0.600	1.000	4.500	4.500	7.500	
EMCC-L76-LL-01	CORRIDOR SMOKE EXHAUST FAN-TWR-LR-SEF-05	2.500	2.500	2.500	7.500	0.800	0.800	0.800	6.000	6.000	6.000	
EMCC-L76-LL-01	CORRIDOR SMOKE MAKE UP FAN-TWR-L77-SMAF-06	0.500	0.500	0.500	1.500	0.800	0.800	0.800	1.200	1.200	1.200	
EMCC-L76-LL-01	CORRIDOR SMOKE EXHAUST FAN-TWR-L77-SEF-06	0.500	0.500	0.500	1.500	0.800	0.800	0.800	1.200	1.200	1.200	
ESMDB-L76-LL-01	CP-WATER BOOSTER PUMP -L76	2.000	2.000	2.000	6.000	0.800	0.800	0.800	2.400	2.400	2.400	1D+1S
ESMDB-L76-LL-01	GSM ISOLATOR-02-GEM RM - ROOF	5.500	5.500	5.500	16.500	0.800	0.800	0.800	13.200	13.200	13.200	
EMDB-GF-01	ESMDB-L24-LL-01											
ESMDB-L24-LL-01	EDB-L24-LL-01	7.833	6.161	6.931	20.925	0.800	0.800	0.800	16.740	16.740	16.740	
ESMDB-L24-LL-01	EDB-L27-LL-01	4.618	8.018	4.618	17.253	0.800	0.800	0.800	13.802	13.802	13.802	
ESMDB-L24-LL-01	EDB-L30-LL-01	5.218	4.818	7.818	17.853	0.800	0.800	0.800	14.282	14.282	14.282	
ESMDB-L24-LL-01	EDB-L33-LL-01	2.151	2.051	1.951	6.153	0.800	0.800	0.800	4.922	4.922	4.922	
ESMDB-L24-LL-01	EDB-L36-LL-01	6.451	2.151	2.151	10.753	0.800	0.800	0.800	8.602	8.602	8.602	
ESMDB-L24-LL-01	EDB-L39-LL-01	7.684	7.284	7.684	22.653	0.800	0.800	0.800	18.122	18.122	18.122	
ESMDB-L24-LL-01	EDB-L42-LL-01	2.251	4.951	1.951	9.153	0.800	0.800	0.800	7.322	7.322	7.322	
ESMDB-L24-LL-01	EDB-L51-LL-01	8.186	7.342	5.937	21.465	0.800	0.800	0.800	17.172	17.172	17.172	
ESMDB-L24-LL-01	EMCC-L24-LL-01											
EMCC-L24-LL-01	STAIRCASE-2 PRESSURISATION FAN-L24TWR-L10-SMAF-04	2.500	2.500	2.500	7.500	0.600	0.600	1.000	4.500	4.500	7.500	
EMCC-L24-LL-01	CORRIDOR SMOKE FAN - TWR-L25-SMAF-03	0.500	0.500	0.500	1.500	0.800	0.800	0.800	1.200	1.200	1.200	
EMCC-L24-LL-01	FIREPUMP ROOM MAKE UP FAN - L24TWR-L25-SMAF-03	0.500	0.500	0.500	1.500	0.800	0.800	0.800	1.200	1.200	1.200	
EMCC-L24-LL-01	FIREPUMPROOM EXHAUST FAN -TWR-L25-SEF-02	0.500	0.500	0.500	1.500	0.800	0.800	0.800	1.200	1.200	1.200	
ESMDB-L24-LL-01	CP-DOMESTIC WATER TRANSFER PUMP-L24	6.000	6.000	6.000	18.000	0.800	0.800	0.800	7.200	7.200	7.200	1D+1S
ESMDB-L24-LL-01	FIRE DIESEL PUMP CONTROL SUPPLY -L24	-	-	0.500	0.500	0.800	0.800	0.800	0.400	0.400	0.400	
ESMDB-L24-LL-01	CP-FIRE JOCKEY PUMP-L24	3.667	3.667	3.667	11.000	0.800	0.800	0.800	8.800	8.800	8.800	
EMDB-GF-01	ESMDB-L52-LL-01								-	-	-	
ESMDB-L52-LL-01	EDB-L54-LL-01	1.841	4.841	1.841	8.523	0.800	0.800	0.800	6.818	6.818	6.818	
ESMDB-L52-LL-01	EDB-L57-LL-01	4.608	4.908	4.408	13.923	0.800	0.800	0.800	11.138	11.138	11.138	
ESMDB-L52-LL-01	EDB-L60-LL-01	4.808	4.408	7.408	16.623	0.800	0.800	0.800	13.298	13.298	13.298	
ESMDB-L52-LL-01	EDB-L63-LL-01	2.041	1.941	3.241	7.223	0.800	0.800	0.800	5.778	5.778	5.778	
ESMDB-L52-LL-01	EDB-L66-LL-01	4.741	1.941	1.941	8.623	0.800	0.800	0.800	6.898	6.898	6.898	
ESMDB-L52-LL-01	EDB-L69-LL-01	7.474	#####	7.474	25.023	0.800	0.800	0.800	20.018	20.018	20.018	
ESMDB-L52-LL-01	EDB-L72-LL-01	2.041	1.741	3.741	7.523	0.800	0.800	0.800	6.018	6.018	6.018	
ESMDB-L52-LL-01	EDB-L75-LL-01	7.229	7.952	5.272	20.453	0.800	0.800	0.800	16.362	16.362	16.362	
ESMDB-L52-LL-01	EMCC-L52-LL-01											
EMCC-L52-LL-01	STAIRCASE-2 PRESSURISATION FAN-TWR-L53-SPF-05	2.500	2.500	2.500	7.500	0.600	0.600	1.000	4.500	4.500	7.500	
EMCC-L52-LL-01	STAIRCASE-3 PRESSURISATION FAN-TWR-L53-SPF-04	2.500	2.500	2.500	7.500	0.600	0.600	1.000	4.500	4.500	7.500	
EMCC-L52-LL-01	CORRIDOR SMOKE FAN - TWR-L53-SEF-04	2.000	2.000	2.000	6.000	0.800	0.800	0.800	4.800	4.800	4.800	

Page 1

ESMDB-B1-LL-01	MTR RACK-01	1.500	1.500	1.500	4.500	0.800	0.800	0.800	3.600	3.600	3.600	
ESMDB-B1-LL-01	MTR RACK-02	1.500	1.500	1.500	4.500	0.800	0.800	0.800	3.600	3.600	3.600	
ESMDB-B1-LL-01	GSM RACK-01	3.000	3.000	3.000	9.000	0.800	0.800	0.800	7.200	7.200	7.200	
ESMDB-B1-LL-01	GSM RACK-02	3.000	3.000	3.000	9.000	0.800	0.800	0.800	7.200	7.200	7.200	
					2,127.854					1,177.483	1,117.683	
					2,127.854							
								TOTAL KW		1177.483	1117.683	

LOAD ON GENERATOR	1177	KW	HIGHEST OF THE TWO (POWER FAILURE, FIRE CONDITION) CONSIDERED
SPARE CAPACITY(10%)	118	KW	
GENERATOR RATING REQUIRED (KW)	1295	KW	
GENERATOR RATING REQUIRED (KVA)	1619	KVA	
GENERATOR RATING SELECTED (KVA)	1750	KVA	

- **This will be continuous sheet listing down all the equipment connected to each panel downstream of the generator;**
- **Users should be able to edit the demand factors for Normal case, power failure case & Fire case for each load**
- **After the demand factors are plugged in, the tool should be able to choose the max demand between case-2 & 3 and propose the generator rating;**

QA Checks

- **For mech equipment, need to include category as “FLS” & “non-FLS” (FLS stands for Fire life safety);**

- **Cables connecting life safety equipment shall be selected automatically as fire rated cables, starting from the equipment and upstream until the generator (this is to ensure during fire these equipment will function and the cables shouldn’t fail) – when a user connects a FLS equipment to a panel, then automatically that cable will be changed to Fire rated but the cables upstream of the panel could have been normal cables because when the user would have selected that cable, at that time there wont have been any FLS equipment downstream; So, the tool should raise a flag to change the upstream cables to fire rated or change it and then inform the user that those cables are changed to fire rated; if this is made automatic, then if the user removes the FLS equipment and feeds it from some other panel, these cables should be automatically changed to normal cable;**

- **Similar to above, if the user has selected fire rated cable and there are no FLS equipment downstream, that should also raise a flag;**

- Total load of the project should be constantly checked against the allocated load for the plot; - this is already added but seems it is now comparing against the load added to one transformer – pls ensure this should be checked against loads connected to all the transformers;

- Total load on each transformer, as loads are being added, should be constantly checked against its capacity;

- Any equipment added by mech & phe engr should be verified that they are connected to some panels; the catch is these equipment can be in different files or layers within the same file;

Back-end Process

Total connected load calculation

- This should be an accumulation of loads from each of panels/isolators/equipment downstream;
- In parallel, the load should be accumulated category-wise at each panels upto the source
  - First categorization - lighting, small power, hvac, water heater, etc;
  - Second categorization – apartment, office, retail, landlord, etc;
  - Third categorization – normal, essential, critical, FLS;

Maximum demand load calculation

- For demand load calculation – there are multiple ways to calculate;
  - For Dubai projects – Demand load is calculated only at the busbar & LV panel level –> for non-Emaar projects 0.65 for residential load, 0.8 for landlord loads, 0.8 for retail/office/hotel loads;
    for Emaar projects, 0.6 for residential load, 0.75 for landlord loads, 0.8 for retail/office/hotel loads;
  - For non-Emaar projects, SMDB's and downstream, Demand factor will be 100%; for Emaar projects, 0.8 demand factor will be applied on SMDB load as well, anything downstream of SMDB will have a demand factor of 100%;

Why do we calculate Demand Load

- Demand load is the basis for calculating the load current and thereby the basis for protective device rating, cable ampacity and voltage drop;